

For **Task 2**,

I need to create a **Smart Agriculture Dashboard Proposal**. Think of it as designing a dashboard that a farmer would use to monitor their farm, crops, weather, and irrigation in one place.

Step-by-Step Guide

Step 1: Create a Title Page

Include:

- KhetHub Internship Program 2026
 - Task 2: Smart Agriculture Dashboard Challenge
 - Submitted by: Pranshu Tyagi
 - Role (Web Development Intern / Backend Developer)
 - Submission Date: 24 June 2026
-

Step 2: Write an Introduction.

Explain what the dashboard does.

Example:

The KhetHub Smart Agriculture Dashboard is a centralized platform that helps farmers monitor crop health, weather conditions, soil moisture levels, irrigation status, and farming activities. The dashboard provides real-time insights and recommendations to improve agricultural productivity.

Step 3: Dashboard Layout Structure

Draw a simple dashboard layout (can be made in Word, PowerPoint, Canva, or Figma).

Sections to Include:

1. Farm Overview

Display:

- Total Farms
- Active Crops
- Total Area
- Irrigation Status

2. Weather Information Panel

Display:

- Temperature
- Humidity

- Rain Forecast
- Wind Speed

3. Crop Health Status

Display:

- Healthy Crops
- Warning Status
- Disease Detection Alerts

4. Soil Moisture Monitoring

Display:

- Current Moisture Level
- Moisture Trend Graph

5. Irrigation Status

Display:

- Pump Status
- Water Usage
- Irrigation Schedule

6. Alerts & Notifications

Display:

- Weather Alerts
- Disease Alerts
- Low Soil Moisture Alerts

7. Task Summary

Display:

- Today's Tasks
- Upcoming Activities
- Completed Tasks

Step 4: Data Visualization

Explain how data will be shown.

Dashboard Cards

Example:

Metric	Value
--------	-------

Temperature	32°C
-------------	------

Humidity	70%
----------	-----

Soil Moisture	55%
---------------	-----

Active Crops	4
--------------	---

Charts

Line Chart

- Soil Moisture Trends

Bar Chart

- Weekly Water Usage

Pie Chart

- Crop Health Distribution

Table

- Recent Farm Activities
-

Step 5: User Experience & Design

Theme

Smart Agriculture Dashboard 

Colors

- Green (#2E7D32)
- Light Green (#81C784)
- White (#FFFFFF)
- Dark Gray (#263238)

Fonts

- Poppins
- Roboto

Design Principles

- Mobile Responsive
- Easy Navigation
- Clear Data Visualization

- Large Readable Text
 - Simple Dashboard Cards
-

Step 6: Technical Approach

For Web Development Intern

Frontend

- HTML
- CSS
- JavaScript
- React.js

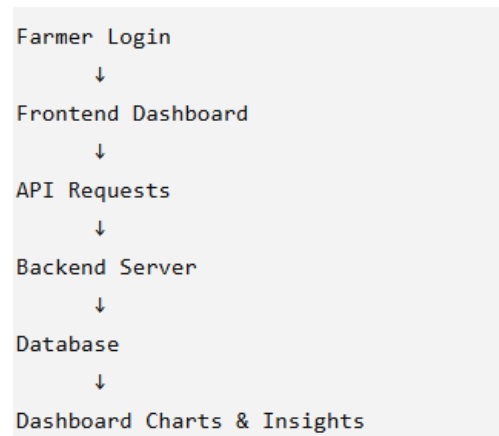
Backend

- Node.js
- Express.js

Database

- MySQL

Workflow



Step 7: Backend Design

Database Schema

Users Table

Field	Type
-------	------

user_id	INT
---------	-----

name	VARCHAR
------	---------

email	VARCHAR
-------	---------

Farms Table

Field	Type
-------	------

farm_id	INT
---------	-----

user_id	INT
---------	-----

farm_name	VARCHAR
-----------	---------

Sensor Data Table

Field	Type
-------	------

sensor_id	INT
-----------	-----

moisture	FLOAT
----------	-------

temperature	FLOAT
-------------	-------

humidity	FLOAT
----------	-------

Alerts Table

Field	Type
-------	------

alert_id	INT
----------	-----

message	TEXT
---------	------

date	DATETIME
------	----------

Step 8: Sample API

Weather API

```
</> JSON

{
  "temperature": 32,
  "humidity": 70,
  "rainForecast": "20%"
}
```

Soil Data API

```
</> JSON

{
  "soilMoisture": 55,
  "status": "Good"
}
```

Crop Health API

```
</> JSON

{
  "healthy": 85,
  "warning": 10,
  "critical": 5
}
```

Step 9: Bonus Features

Add these to score higher:

- Dark Mode
- AI Crop Recommendations
- Real-Time Sensor Monitoring
- Multilingual Support (Hindi & English)
- Mobile-Friendly Design
- Interactive Charts

Step 10: Conclusion

Example:

The KhetHub Smart Agriculture Dashboard will provide farmers with real-time agricultural insights through weather monitoring, crop health tracking, soil moisture analysis, irrigation management, and intelligent alerts. The platform will improve decision-making, increase productivity, and support modern digital farming practices.

Khethub Smart Agriculture dashboard

 KhethHub Smart Agriculture Dashboard

Temperature
32°C

Humidity
70%

Soil Moisture
55%

Active Crops
4

 Weather Information

Sunny | Rain Chance: 20% | Wind Speed: 12 km/h

 Crop Health Status

Healthy: 85%
Warning: 10%
Critical: 5%

 Irrigation Status

Pump Status: ON
Water Usage: 120 Litres

 Alerts

- Low Soil Moisture Alert
- Rain Expected Tomorrow

 Today's Tasks

- Water Field A
- Check Crop Health
- Apply Fertilizer

Code 1: Index.HTML

```
Welcome  index.html1.html X JS script.js # style.css 1.css
index.html1.html > html > head > link
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>KhetHub Smart Agriculture Dashboard</title>
7   <link rel="stylesheet" href="style.css">
8 </head>
9 <body>
10
11 <header>
12   <h1> KhetHub Smart Agriculture Dashboard</h1>
13 </header>
14
15 <div class="container">
16
17   <div class="cards">
18     <div class="card">
19       <h3>Temperature</h3>
20       <p id="temp">32°C</p>
21     </div>
22
23     <div class="card">
24       <h3>Humidity</h3>
25       <p id="humidity">70%</p>
26     </div>
27
28     <div class="card">
29       <h3>Soil Moisture</h3>
30       <p id="moisture">55%</p>
31     </div>
32
```

Code 2: style.css

```
Welcome  index.html1.html JS script.js # style.css 1.css X
# style.css 1.css > .card h3
1 *{
2   margin:0;
3   padding:0;
4   box-sizing:border-box;
5   font-family:Arial, sans-serif;
6 }
7
8 body{
9   background: #f4f7f4;
10 }
11
12 header{
13   background: #2e7d32;
14   color: white;
15   text-align:center;
16   padding:20px;
17 }
18
19 .container{
20   width:90%;
21   margin:20px auto;
22 }
23
24 .card{
25   background: white;
26   padding:10px;
27   margin-bottom:10px;
28 }
29
30 .card h3{
31   margin:0;
32 }
33
34 .card p{
35   margin:0;
36 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, lexcludeText, ...) Prettier

["INFO" - 10:42:46 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

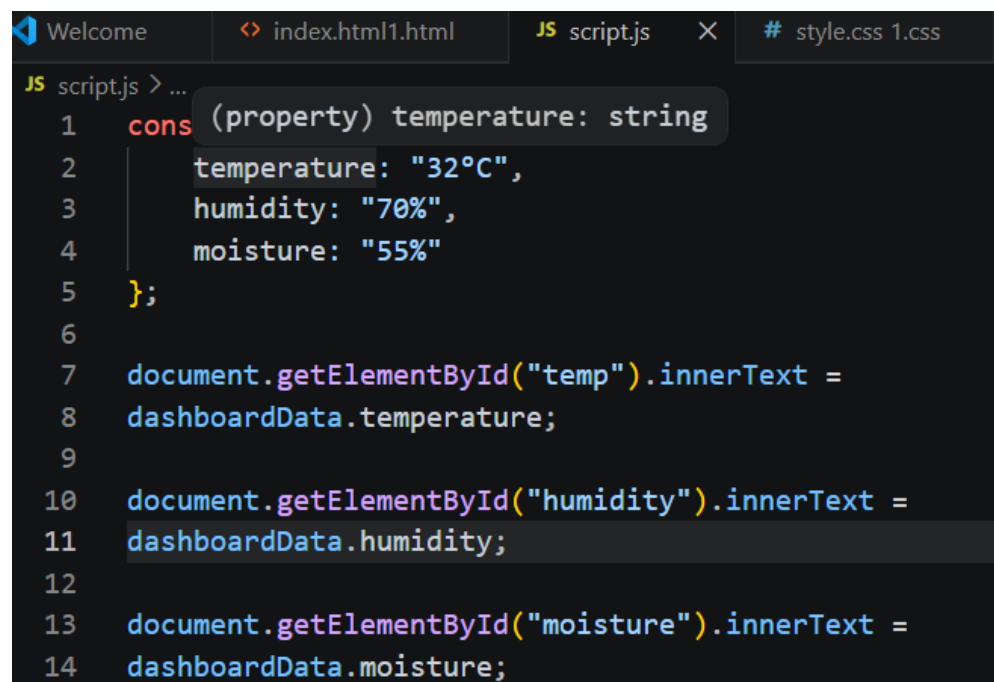
["INFO" - 10:52:31 PM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 10:52:31 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

["INFO" - 10:53:12 PM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 10:53:12 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

Code 3: script.js

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'Welcome', 'index.html1.html', and 'script.js' (which is active). Below the tabs, the code in 'script.js' is displayed with line numbers 1 through 14. The code defines a 'dashboardData' object with 'temperature', 'humidity', and 'moisture' properties, and then uses 'document.getElementById' to set the 'innerText' of elements with IDs 'temp', 'humidity', and 'moisture' to the corresponding values in the object. A tooltip is visible over the 'temperature' property in the object definition, showing '(property) temperature: string'.

```
JS script.js > ...
1  cons (property) temperature: string
2      temperature: "32°C",
3      humidity: "70%",
4      moisture: "55%"
5  };
6
7  document.getElementById("temp").innerText =
8  dashboardData.temperature;
9
10 document.getElementById("humidity").innerText =
11 dashboardData.humidity;
12
13 document.getElementById("moisture").innerText =
14 dashboardData.moisture;
```